ECON 101 End Term Exam

Question Paper

100 Marks (3 Hours)

Question 1 [20 Marks] – There are two individuals (A and B) who work in the same firm. Profits depend upon the joint efforts that these individuals decide to put into their work and all the profits are divided equally between the two individuals. Both individuals can either choose to devote high effort or low effort. If an individual decides to devote high effort then it costs that individual 50 while low effort costs only 10. If both of them decide to devote high effort then the firm earns a profit of 1000, while if both of them decide to devote low effort then the firm earns a profit equal to 500. If one individual decides to put in high effort and the other decides to put in low effort then the firm earns a profit of 950.

a) Write down the Normal Form Representation of this game. (5 Marks)

b) Are there any dominant strategies in this game, find the dominant strategy equilibrium? (5 Marks)

c) For each of the strategy combinations verify if that combination constitutes Nash equilibrium. (5 Marks)

d) If you think that a particular strategy combination in (c) constitutes a Nash Equilibrium, explain. (5 Marks)

Question 2 [20 Marks] – A firm produces output y using inputs x_1 and x_2. In order to do so it uses the following production technology: $y = 2x_1 + 3x_2$. Suppose that the price of input x_1 is 5 while the price of input x_2 is 10.

a) Assuming that the firm wants to produce y_0 level of output, state the cost minimization problem for the firm (5 Marks).

b) Use this information in order to solve for the cost minimizing level of inputs needed to produce y_0 units of good, describe the same using a well labelled diagram (5 Marks).

c) Solve for the cost function for this firm (10 marks).

Question 3 [20 Marks] – A firm has the following cost function: $C(Q) = Q^3 - 20Q^2 + 50Q + 100$. Use this information to answer the following –

a) Calculate the following – Marginal Cost, Average Cost, Fixed Cost, Average Variable Cost, Average Fixed Cost (5 Marks)

b) Use the above calculated values of average variable cost and marginal cost in order to describe the relationship between AVC and MC. Plot the above calculated AVC and MC on a well labelled diagram as well (7+3 Marks).

c) The long run average cost curve lies above the various short run average cost curves. Do you agree with this statement? Explain. (5 Marks)

Question 4 [10 Marks] There are four individuals X, Y, A and B. X and Y **each** have one unit of an indivisible good, and no money. The value they attach to one unit of good is as follows - \$0 for X and \$3 for Y. A and B **each** have \$5 in money and no units of the good. The value they attach to one unit of good is as follows - \$5 for A and \$2 for B. No person has any value for a second unit of the good. The payoff of a person is defined as the value that person places on the unit of the good plus the money he/she has. Use this information in order to answer the following –

a) Explain whether the initial distribution of the good and money is pareto efficient or not. (5 Marks)

b) Suppose that X decided to sell her unit of good to B for \$2, Y sells her unit to A for \$3; explain whether this new distribution of good and money is a pareto efficient allocation or not. (5 Marks)

Question 6 [10 Marks] A monopolist producer faces a constant elasticity market demand curve, wherein the price elasticity of demand equals (-2). Also, the monopolist produces output at a constant marginal cost of 10 per unit. Use this information in order to calculate the price charged by the monopolist.

Question 7 [20 Marks] A consumer consumes two good x and y and the following utility function describes the preferences for the consumer - $U(x, y) = min\{2x + 3y, 3x + 2y\}$. The consumer has an income of 100, which is spent on consuming x and y. Price of good x is 5 while the price of good y is 10. Graph the indifference curves and budget line for the consumer. Use the same in order to solve for the chosen bundle and explain as well.